

LUCIDPREP

Grade 7 Life Science

U.S. Standards-Aligned Course Syllabus

LIFE SCIENCE

Life Science • NGSS

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

There is no single national science curriculum, but the Next Generation Science Standards (NGSS) are the dominant U.S. reference. NGSS middle-school standards are written as a 6–8 band rather than by single grade; this syllabus places Life Science at Grade 7, a common domain-based arrangement. Districts using integrated middle-school sequences will redistribute these standards across grades 6–8.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

LIFE SCIENCE

Grade 7 Life Science aligned to
the Next Generation Science
Standards

This 36-week course is designed around **the NGSS Middle School Life Science standards (MS-LS1-MS-LS4), with Science & Engineering Practices and Crosscutting Concepts**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 7
Course	Life Science · NGSS
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Explain that living things are made of cells and how structures support function.
- Describe how organisms grow, develop and reproduce.
- Trace matter and energy through organisms and ecosystems.
- Analyze interactions, dynamics and human impact within ecosystems.
- Explain natural selection, adaptation and evidence of common ancestry.

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Module 1 — Cells & Body Systems

Week	Topics	Students will be able to	Standards
1 — Characteristics of Life & Investigation	Living vs nonliving, scientific method, tools	Plan investigations and distinguish living things	MS-LS1-1
2 — Cells & Cell Theory	Cells as the units of life, single/multicellular	Provide evidence that living things are made of cells	MS-LS1-1
3 — Cell Structure & Function	Organelles, plant vs animal cells	Model how cell parts contribute to function	MS-LS1-2
4 — From Cells to Organisms	Cells → tissues → organs → systems	Explain hierarchical organization	MS-LS1-3
5 — Body Systems I	Digestive, circulatory, respiratory interactions	Explain how body systems interact	MS-LS1-3
6 — Body Systems II	Nervous system, sensory receptors, response	Explain sensing and response to stimuli	MS-LS1-8
7 — Homeostasis	Maintaining stable internal conditions	Explain how organisms maintain homeostasis	MS-LS1-3
8 — Cells & Health	Investigating cell processes and applications	Apply cell concepts to real phenomena	MS-LS1
9 — Module 1 Assessment	Cumulative review of cells and body systems	Demonstrate mastery of Module 1	MS-LS1

Module delivery notes. Use models, investigations and data analysis to make sense of living systems. NGSS middle-school standards are banded 6–8; districts arrange them differently, so pacing may shift.

Module 2 — Matter, Energy & Growth

Week	Topics	Students will be able to	Standards
1 — Photosynthesis	How plants make food from light	Model how plants use light to make food	MS-LS1-6
2 — Cellular Respiration & Food as Energy	Chemical energy stored in food	Describe how organisms use food for energy	MS-LS1-7
3 — Matter & Energy Flow in Organisms	Rearrangement of atoms and molecules	Explain matter and energy flow in organisms	MS-LS1-7
4 — Growth & Development	How organisms grow; factors affecting growth	Explain organism growth and development	MS-LS1-5
5 — Reproduction & Success	Sexual/asexual reproduction, characteristic behaviors	Explain how reproduction affects success	MS-LS1-4
6 — Genes, Traits & Inheritance	Chromosomes, genes, proteins	Explain how genes determine traits	MS-LS3-1
7 — Variation & Mutations	Causes of variation; effects of mutations	Explain sources of genetic variation	MS-LS3-1
8 — Heredity & Probability	Predicting inheritance and offspring variation	Predict patterns of inheritance	MS-LS3-2
9 — Module 2 Assessment	Cumulative review of matter, energy and heredity	Demonstrate mastery of Module 2	MS-LS1/LS3

Module delivery notes. Use models, investigations and data analysis to make sense of living systems. NGSS middle-school standards are banded 6–8; districts arrange them differently, so pacing may shift.

Module 3 — Ecosystems & Interactions

Week	Topics	Students will be able to	Standards
1 — Ecosystem Organization	Organisms, populations, communities, ecosystems	Describe levels of organization	MS-LS2-1
2 — Resources & Populations	Resource availability and population effects	Analyze effects of resources on populations	MS-LS2-1
3 — Interactions Among Organisms	Competition, predation, symbiosis	Classify and predict interactions	MS-LS2-2
4 — Energy Flow	Food webs, trophic levels, energy pyramids	Trace energy through ecosystems	MS-LS2-3
5 — Cycling of Matter	Carbon, water and nutrient cycles; decomposers	Model the cycling of matter	MS-LS2-3
6 — Ecosystem Stability & Change	Disruptions and resilience	Evaluate effects of change on ecosystems	MS-LS2-4
7 — Biodiversity & Ecosystem Services	Importance of biodiversity	Evaluate design solutions for biodiversity	MS-LS2-5
8 — Human Impact on Ecosystems	Pollution, habitat change, solutions	Analyze and propose solutions to human impact	MS-ESS3-3/4
9 — Module 3 Assessment	Cumulative review of ecosystems	Demonstrate mastery of Module 3	MS-LS2

Module delivery notes. Use models, investigations and data analysis to make sense of living systems. NGSS middle-school standards are banded 6–8; districts arrange them differently, so pacing may shift.

Module 4 — Evolution, Diversity & Adaptation

Week	Topics	Students will be able to	Standards
1 — Fossil Evidence	The fossil record and patterns over time	Analyze fossil-record patterns	MS-LS4-1
2 — Anatomical Evidence	Similarities/differences and common ancestry	Use anatomy to infer relationships	MS-LS4-2
3 — Embryological & Molecular Evidence	Developmental and molecular similarities	Use additional evidence for ancestry	MS-LS4-3
4 — Natural Selection	Variation, differential survival and reproduction	Explain natural selection	MS-LS4-4
5 — Adaptation Over Time	Trait change in populations	Use evidence to explain adaptation	MS-LS4-6
6 — Artificial Selection	Selective breeding by humans	Explain how humans influence inheritance	MS-LS4-5
7 — Classification & Diversity	Grouping organisms; biodiversity	Classify and compare organisms	MS-LS4
8 — Evolution in Action	Real-world examples (e.g., resistance)	Analyze evidence of ongoing evolution	MS-LS4-4/6
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	MS-LS1-LS4

Module delivery notes. Use models, investigations and data analysis to make sense of living systems. NGSS middle-school standards are banded 6-8; districts arrange them differently, so pacing may shift.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	NGSS: MS-LS1, MS-LS2, MS-LS3, MS-LS4
Science & Engineering Practices	Asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; obtaining, evaluating and communicating information.
Crosscutting Concepts	Patterns; cause and effect; scale, proportion and quantity; systems and system models; energy and matter; structure and function; stability and change.
State mapping	NGSS adopted or adapted by nearly all states; middle-school standards are banded 6–8 and arranged differently by state.

Content Production Guidance

Author each week around an anchoring phenomenon or problem: 5 warm-ups, an explanatory lecture with models, 2 videos, an AI presentation with visual models and investigations, 2 homework sets, and a 40-question bank mixing three-dimensional items (SEP × DCI × CCC). Tag items to NGSS performance expectations.

Curriculum Basis & Source Notes

This Grade 7 science syllabus was generated from the Next Generation Science Standards. It reflects the Life Science · NGSS course; because NGSS middle-school standards are banded 6–8, the arrangement here is one common domain-based model.

Primary reference	Next Generation Science Standards — MS (grades 6–8) performance expectations, with Science & Engineering Practices and Crosscutting Concepts.
Framework basis	A Framework for K-12 Science Education (National Research Council).
State cross-check	California and other NGSS-adopting state science frameworks.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Select anchoring phenomena per module and commission three-dimensional item banks (SEP × DCI × CCC) tagged to NGSS performance expectations.